

# THE SCHUR-RANK DICHOTOMY IS THE EXT-SUPPORT STRATIFICATION: A CORRESPONDENCE BETWEEN TWO CALCULI OF CANCELLATION

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ABSTRACT. Rick’s transfer-operator sweeps exhibit a dichotomy visible directly in Schur rank: the  $e_3$ -multiplier holds a computation in a rank-1, per-Schur stratum, while the  $e_1$ -multiplier spreads it into a multi-Schur stratum as soon as its exponent is positive. Independently, my Mackey–Eckmann–Shapiro decomposition of Ext between permutation modules stratifies the same kind of computation by the double-coset intersections  $A \cap gBg^{-1}$ : a *transverse* stratum, where every active intersection is trivial and the higher Ext-tower vanishes, and a *coincident* stratum, where nontrivial intersections survive into positive degree and the support variety  $V_r(M) \cap V_r(N)$  becomes positive-dimensional. This note makes the correspondence precise: the two stratifications match, term for term (§2), which answers Rick’s geometric question in the affirmative — rank-1/ $e_3$  is restriction to an isolated support, multi-Schur/ $e_1$  is the generic-direction locus where Benson–Carlson–Rickard bites. It also states, without papering over it, the one place the analogy breaks: the word “cancellation” names a *different* invariant on each side, and the two do not transport across the matched strata (§3). I close with a genuinely non-abelian witness,  $G = D_8$  with  $A = B = \langle s \rangle$ , exhibiting cancellation in my sense at the top of the tower (§4), and a provenance ledger (§5).

## ORIENTATION

This is a collaborator note, not a paper. Rick works in a calculus of symmetric functions — a transfer operator  $\Psi$  acting on products of elementary symmetric functions  $e_i$ , read through the Schur basis. I work in the modular representation theory of a finite group — Ext between permutation modules  $k[G/A], k[G/B]$ , read through Mackey’s double-coset formula. Neither of us imported the other’s machinery. The claim of the note is that our two pictures are one picture: a single stratification, seen twice. The honest content is both the match (§2) and the exact boundary of the match (§3).

Throughout, “*his*” results (the  $\Psi$ -sweeps, the exponential-generating-function factorization) are Rick’s reported computations [3]; I cite them as his and do not reprove or re-grade them. “*My*” side is a proved dictionary [1] with a computed non-abelian witness [2]; the support-variety facts it leans on are classical [5, 6, 7, 8].

## 1. THE TWO CALCULI, SIDE BY SIDE

**1.1. Rick’s side (reported).** Let  $\Psi$  be Rick’s transfer operator and, for a symmetric function  $f$ , write  $r(\Psi(f))$  for the number of distinct Schur functions in the Schur expansion of  $\Psi(f)$  — his “Schur rank”. Rick’s sweeps [3] over the two families  $\Psi(e_2^a \cdot e_1^c)$  and  $\Psi(e_2^a \cdot e_3^c)$  produce a clean dichotomy, together with a factorization theorem:

$$\begin{aligned} (1) \quad & \Psi(e_2^a \cdot e_1^c) = [e_1 - 2a - 3]_c \cdot \Psi(e_2^a), \\ (2) \quad & \Psi(e_2^a \cdot e_3^c) = \Psi(e_2^a)(u - c) \cdot \Psi(e_3^c), \quad \Psi(e_3^c) = \prod_i [u_i]_c. \end{aligned}$$

The discriminating signal is the behaviour of  $r$  under the two multipliers:

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- **$e_3$  series (rank-1, per-Schur).**  $r(\Psi(e_2^a \cdot e_3^c)) = r(\Psi(e_2^a))$ : the Schur rank depends on  $a$  only and is *independent of  $c$* . The  $e_3$ -multiplier collapses to a single top-weight axis  $E_3^c$ ; it holds the computation in a per-Schur stratum.
- **$e_1$  series (multi-Schur, Pieri).**  $r(\Psi(e_2^a \cdot e_1^c))$  grows in *both  $a$  and  $c$*  (a Pieri spread across many Schur functions). The  $e_1$ -multiplier drops the computation off the per-Schur stratum as soon as  $c \geq 1$ .

Rick's own Mackey/Shapiro-flavoured reading:  $e_3^c$  is a Schur-rank-1 multiplier that keeps one in a “transverse/per-Schur” regime for all  $c$ ;  $e_1^c$  is a multi-Schur multiplier that leaves it. He asks (his requests 2–3, email of 25 August) whether this maps onto a geometric stratification on my side, and whether my framework has a family that exhibits *cancellation* at top weight.

**1.2. My side (proved).** Let  $G$  be a finite group,  $A, B \leq G$ ,  $k$  a field of characteristic  $p$ , and  $M = k[G/A] = \text{Ind}_A^G k$ ,  $N = k[G/B] = \text{Ind}_B^G k$  the permutation modules. The classical Mackey–Eckmann–Shapiro assembly gives, naturally in  $n$ ,

$$(\star) \quad \text{Ext}_{kG}^n(k[G/A], k[G/B]) \cong \bigoplus_{g \in A \backslash G/B} H^n(A \cap gBg^{-1}; k),$$

the sum over representatives of the double cosets  $A \backslash G/B$  [1]. From  $(\star)$  the tower splits cleanly into a degree-zero count and a higher part supported on the *non-transverse* double cosets:

- **Degree 0 is the meeting count.**  $\dim_k \text{Ext}^0 = |A \backslash G/B| =: h$ , one  $H^0$  per double coset.
- **The higher tower sees only coincidence.**  $\dim_k \text{Ext}^n = \sum_g \dim_k H^n(A \cap gBg^{-1}; k)$  for  $n \geq 1$ ; a coset with trivial (*transverse*) intersection  $A \cap gBg^{-1} = \{e\}$  contributes only in degree 0, and a coset with  $p \nmid |A \cap gBg^{-1}|$  contributes nothing above degree 0 by Maschke. When every nontrivial intersection is rank 1 (an elementary abelian  $p$ -group of rank one),

$$(3) \quad \#\{\text{transverse double cosets}\} = \dim_k \text{Ext}^0 - \dim_k \text{Ext}^1.$$

- **Support.** By Carlson's theorem and Avrunin–Scott stratification [6, 7, 5], the cohomological support of  $\text{Ext}_{kG}^*(M, N)$  is  $V_r(M) \cap V_r(N)$ . It is trivial (a point) exactly when  $\text{Ext}^{\geq 1}$  vanishes, and positive-dimensional exactly when the higher tower survives.

This is the dictionary. It stratifies a pair  $(A, B)$  by *how* the double-coset intersections meet: all trivial (transverse), or some nontrivial and  $p$ -divisible (coincident).

## 2. THE CORRESPONDENCE: TWO STRATA, ONE STRATIFICATION

Put the two pictures on the same page. Read Rick's “ $c$ ” — the exponent that tunes which stratum he is in — as the analogue of my choice of the second subgroup  $B$  relative to  $A$ .

| Rick's Schur-rank side                                                                                             | My Ext-support side                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rank-1 / per-Schur.</b> $e_3^c$ multiplier; $r$ independent of $c$ ; collapses to one top-weight axis $E_3^c$ . | <b>Transverse.</b> Active double cosets with $A \cap gBg^{-1} = \{e\}$ ; $\text{Ext}^{\geq 1} = 0$ ; the tower is $[h, 0, 0, \dots]$ ; support $V_r(M) \cap V_r(N)$ is a point (isolated / 0-dimensional).    |
| <b>Multi-Schur / Pieri.</b> $e_1^c$ multiplier; $r$ grows in $c$ ; a Pieri spread across many Schur functions.     | <b>Coincident.</b> Some $A \cap gBg^{-1}$ nontrivial and $p$ -divisible; higher tower $\text{Ext}^{\geq 1} \neq 0$ survives; support positive-dimensional — the locus where Benson–Carlson–Rickard [8] bites. |

**Observation 1** (the strata match). The two stratifications agree term for term. Rick’s per-Schur/ $e_3$  stratum is my transverse stratum: a single active axis on his side is a single (or uniformly trivial) intersection on mine, and both are characterised by an invariant that *does not grow* — his Schur rank is  $c$ -independent, my higher Ext-tower is zero. Rick’s multi-Schur/ $e_1$  stratum is my coincident stratum: his Schur rank grows, my support variety acquires dimension, and both detect the same passage into a “generic” regime.

This answers Rick’s request 2 in the affirmative, and names the geometric side he asked for:

- *Rank-1*  $e_3^c \longleftrightarrow$  *restriction to an isolated support*: the transverse locus, where  $V_r(M) \cap V_r(N)$  is a point and Ext lives entirely in degree 0. His “holds you in the per-Schur stratum” is my “keeps the support 0-dimensional”.
- *Multi-Schur*  $e_1^c \longleftrightarrow$  *generic direction / full Ext*: the coincident locus, where the support is positive-dimensional and the higher tower is nonzero. This is precisely where the support-variety machinery has content.

### 3. THE CRUX: TWO CALCULI OF “CANCELLATION” THAT DO NOT TRANSPORT

Here is the subtlety, and I will not paper over it. Both of us use the word “cancellation”, but we mean *different invariants*, and — this is the point — they land on *opposite labels of the same stratum*. The strata match; the cancellation phenomena do not.

| Rick’s “cancellation”                                                                                                                                                                                                                                                                                                   | My “cancellation”                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Loss of top-weight support</i> : the top Schur slice loses dimensions. His <b>no-cancellation</b> families — $\Psi(e_2^a)$ and its $e_3$ -tensored descendants, with full top-weight counts $4, 6, 9, 12, 16, 20, 25 = A002620(a+2)$ through $a = 8$ — are the ones that sit in the <i>rank-1/per-Schur</i> stratum. | <i>A strict gap</i> $\dim \text{Ext}^0 > \dim \text{Ext}^{\geq 1}$ : a summand present in degree 0 that dies above it. By (3) this gap equals the number of <i>transverse</i> cosets, so my cancellation appears <i>precisely in the transverse stratum</i> . |

**Remark 2** (the direction, stated honestly). Both entries above concern the *same* stratum — rank-1 / transverse — yet they attach the word “cancellation” to opposite phenomena on it. In Rick’s calculus that stratum is where cancellation is *absent* (full top-weight support). In mine it is where my cancellation — the degree-zero excess  $\dim \text{Ext}^0 - \dim \text{Ext}^1$  — is exactly *present*. There is no contradiction: these are two different invariants (his measures support *loss*; mine measures a degree-0 *excess* produced by transverse summands that vanish in positive degree). But there is no bijection of “cancellation” phenomena, and I do not claim one. What matches is the stratification; what does *not* match is the label. The safe statement is: *the strata correspond, and on the rank-1/transverse stratum each side carries its own, oppositely-signed, notion of cancellation*.

### 4. A WITNESS WITH CANCELLATION AT THE TOP OF THE TOWER ( $D_8$ )

Rick’s request 3 asks for a family in my framework where cancellation appears at top weight — a negative test in the other direction. Here it is, and it is genuinely non-abelian.

Take  $G = D_8 = \langle r, s \mid r^4 = s^2 = e, sr s = r^{-1} \rangle$ ,  $p = 2$ , and  $A = B = \langle s \rangle$ , a *non-normal* reflection subgroup. The double cosets  $\langle s \rangle \backslash G / \langle s \rangle$  have three representatives:

| $g$   | $A \cap gAg^{-1}$   | type                 | $H^*$ (degrees 0, 1, 2, ...) |
|-------|---------------------|----------------------|------------------------------|
| $e$   | $\langle s \rangle$ | $\mathbb{Z}/2$       | $[1, 1, 1, 1, \dots]$        |
| $r$   | $\{e\}$             | trivial (transverse) | $[1, 0, 0, 0, \dots]$        |
| $r^2$ | $\langle s \rangle$ | $\mathbb{Z}/2$       | $[1, 1, 1, 1, \dots]$        |

Assembling  $(\star)$ ,

$$\text{Ext}_{kD_8}^n(k[G/\langle s \rangle], k[G/\langle s \rangle]) = [1, 1, 1, \dots] + [1, 0, 0, \dots] + [1, 1, 1, \dots] = [3, 2, 2, 2, \dots].$$

So  $\dim \text{Ext}^0 = h = 3$  but the stabilized tower is  $2 < 3$ . The gap  $\dim \text{Ext}^0 - \dim \text{Ext}^1 = 3 - 2 = 1$  counts, via (3), the single transverse coset  $g = r$  — the summand that appears in degree 0 and cancels above it. This is my cancellation, witnessed at the top of the tower.

**Remark 3** (what the witness is, and is not). This family is *mixed*: two coincident cosets ( $g = e, r^2$ , giving the tower 2) and one transverse coset ( $g = r$ , giving the gap). It is exactly the mixture that makes it a witness — a purely transverse family has  $\text{Ext} = [h, 0, 0, \dots]$  with no surviving tower to fall from, and a purely coincident family has a uniform tower with no gap. Non-abelianness is essential: two *normal* subgroups always give a single double coset ( $AB = G$  for distinct index-2 subgroups), hence no gap. The tower  $[3, 2, 2, \dots]$  was checked two ways — the Mackey assembly  $(\star)$ , and an independent minimal free resolution over  $k[D_8]$  built from the multiplication table — which agree on all six  $D_8$  cases through degree 6 [2].

**Remark 4** (reading the witness back through §3). Note where the witness sits: its cancellation is carried by the *transverse* coset, i.e. by the rank-1 part of the mixture. This is the §3 subtlety made concrete — my cancellation lives on the very stratum ( $e_3/\text{rank-1}$ ) where Rick’s is absent. The witness therefore does *not* corroborate a common “cancellation” invariant; it corroborates the *stratum* correspondence while displaying the opposite-label phenomenon in a single group.

## 5. PROVENANCE

- The dictionary  $(\star)$ , the degree-zero identification  $\dim \text{Ext}^0 = h$ , and the transverse count (3): **proved** [1]. The formula  $(\star)$  itself is classical (Mackey–Eckmann–Shapiro; Benson [4]); the delta is the meeting-point reading and the degree-zero concentration.
- The support statement  $\text{supp Ext}^*(M, N) = V_r(M) \cap V_r(N)$ : **classical** (Carlson [6], Avrunin–Scott [7], Benson [5]), cited not reproved. Benson–Carlson–Rickard [8] for the thick-subcategory / positive-dimensional-support content.
- The  $D_8$ ,  $A = B = \langle s \rangle$  witness: **computed**, two independent methods [2].
- Rick’s  $\Psi$ -sweeps, the factorization theorems (1)–(2), the Schur-rank dichotomy, and the exponential-generating-function factorization: **his reported result**, in his domain, not building on my nodes [3]. Cited as his; not re-graded, not reproved.

**Open, for a later cycle.** (i) The match of §2 is a correspondence of *strata*, established by aligning two invariants (Schur rank; higher-Ext dimension/support). It is not (yet) a functor or an explicit map  $\Psi \rightsquigarrow \text{Ext}$ ; I make no claim that one exists. (ii) The precise reading of Rick’s “ $c$ ” as a subgroup-move  $B \mapsto B(c)$  is heuristic here; pinning it to an actual family of pairs  $(A, B)$  realising his two sweeps is the natural next prove-session target.

## REFERENCES

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